Joinpoint Regression Melanoma Analysis

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Learning Objectives

- Understand the goals and methods of joinpoint regression.
- Utilize the segmented package in R to run joinpoint regression models.
- Visualize and summarize cancer incidence and trends by sex, race, and cancer subtype.

Code

```r
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.5
✓ forcats   1.0.0     ✓ stringr   1.5.1
✓ ggplot2   4.0.1     ✓ tibble    3.3.0
✓ lubridate 1.9.4     ✓ tidyr     1.3.1
✓ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(segmented)
```

```
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select

Loading required package: nlme

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

    collapse
```

```
set.seed(0)

# Load and clean data
data <- read_csv("Melanoma_5.csv")
```

```
Rows: 3675 Columns: 7
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (5): Race recode (White, Black, Other), Sex, Primary Site - My Variable,...
dbl (2): Count, Population

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
colnames(data) = c("Race", "Sex", "Site", "Year", "Rate", "Count", "Population")

data$Rate <- as.numeric(data$Rate)
```

```
Warning: NAs introduced by coercion
```

```
data$Year = as.numeric(data$Year)
```

```
Warning: NAs introduced by coercion
```

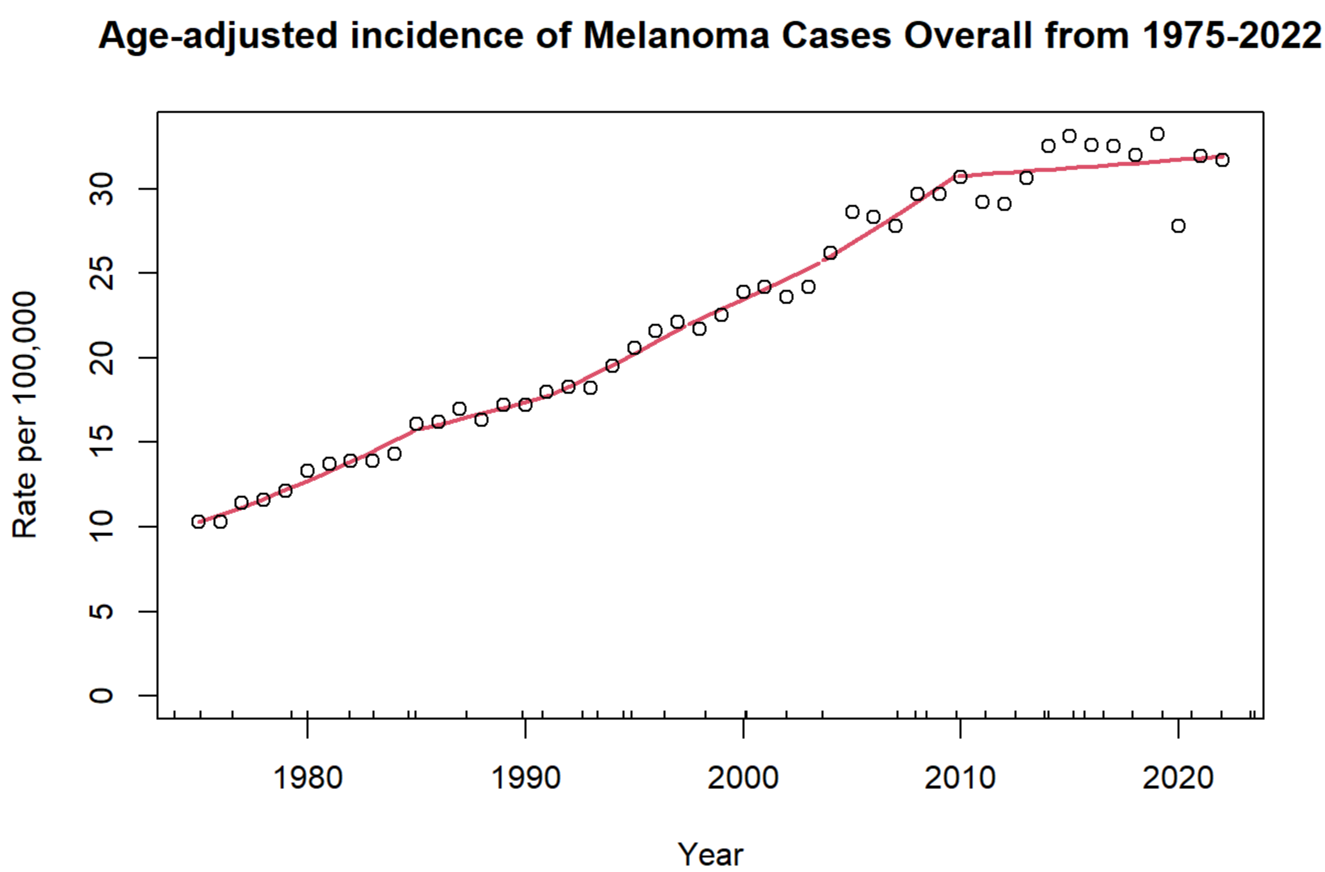
```
# Create age-adjusted counts variable for the data
data$Count_adj = round(data$Rate*data$Population)

# Remove the year range
data_melanoma <- data|> filter(Year != "1975-2022")

# Select a data subset (overall)
data_melanoma_overall = data_melanoma |>
  filter(Race == c("All races") &
           Sex == "Male and female" &
           Site == "All sites")

# Log-linear model (Poisson)
out.glm = glm(Count_adj ~ Year, data= data_melanoma_overall,
              family=poisson(), offset=log(Population))

# Select the best fit number of joinpoints
out.glm.best= selgmented(out.glm, seg.Z = ~ Year, Kmax = 5, type="bic")
```

```
No. of breakpoints: 2 .. 3 .. 4 .. 5 ..
```

```
Warning: The best BIC value at the boundary. Increase 'Kmax'?
```

```
BIC to detect no. of breakpoints:
        0         1         2         3         4         5
142447930  36608524  32741279  32927743  32872455  32028633

No. of selected breakpoints: 5
```

```
plot.segmented(
  out.glm.best,
  link = FALSE,
  ylim = c(0, max(data_melanoma_overall$Rate)),
  main = "Age-adjusted incidence of Melanoma Cases Overall from 1975-2022",
  ylab = "Rate per 100,000")
points(data_melanoma_overall$Year, data_melanoma_overall$Rate)
```




```
# Average annual percent change, 1985-2000
# There is an aapc() function, but doesn't have the flexibility to specify the
# periods of interest. It only calculates it on the full range.
# So, let's calculate by hand.
summary(out.glm.best)
```

```
    ***Regression Model with Segmented Relationship(s)***

Call:
segmented.glm(obj = olm, seg.Z = seg.Z, psi = start.ora, control = control1)

Estimated Break-Point(s):
               Est. St.Err
psi1.Year 1984.924  0.003
psi2.Year 1991.142  0.005
psi3.Year 1997.308  0.007
psi4.Year 2003.458  0.015
psi5.Year 2009.700  0.002

Coefficients of the linear terms:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -8.251e+01  1.701e-02 -4851.4   <2e-16 ***
Year         4.296e-02  8.590e-06  5000.9   <2e-16 ***
U1.Year     -2.326e-02  1.465e-05 -1588.0       NA
U2.Year      1.422e-02  1.772e-05   802.4       NA
U3.Year     -8.390e-03  1.759e-05  -476.9       NA
U4.Year      3.585e-03  1.553e-05   230.8       NA
U5.Year     -2.606e-02  1.063e-05 -2451.6       NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null     deviance: 1929921126  on 47  degrees of freedom
Residual deviance:   32027552  on 36  degrees of freedom
AIC: 32028611

Convergence attained in 23 iterations (rel. change -8.064e-06)
```

```
# The relevant break points at 1984, 1995, and 2005
slope(out.glm.best, APC=TRUE)
```

```
$Year
          Est. CI(95%).l CI(95%).u
slope1 4.38920   4.38740   4.39090
slope2 1.98860   1.98620   1.99100
slope3 3.44880   3.44620   3.45150
slope4 2.58460   2.58220   2.58690
slope5 2.95300   2.95090   2.95510
slope6 0.30478   0.30422   0.30534
```

```
# We see that the relevant beta coefficients are -0.0344310 (for 1985-1995:
# 10 years) and -0.0202100 (for 1995 to 1999: 5 years)
```

```
AAPC_est = (exp((5*0.003 + 5*0.003)/10) - 1)
# AAPC on this 15 year period is -2.38% per year
```

```
joinpoint_model <- function(data_melanoma, race, sex, site, kmax = 5) {

  data_melanoma_filtered <- data_melanoma |>
    filter(Race == race &
             Sex == sex &
             Site == site)

  # Log-linear model (Poisson)
  out.glm <- glm(Count_adj ~ Year, data = data_melanoma_filtered,
      family=poisson(), offset=log(Population))

  # Select the best fit number of joinpoints
  out.glm.best <- selgmented(out.glm, seg.Z = ~ Year, Kmax = kmax, type = "bic", msg = FALSE)
  if (is.null(out.glm.best) || is.null(out.glm.best$psi) || nrow(out.glm.best$psi) == 0) {
    return(NULL)
  }

  # Identify joinpoints
  num_breakpoints <- nrow(out.glm.best$psi)
  joinpoint_years <- round(out.glm.best$psi[, 'Est.'])

  # Identify segment years
  min_year <- min(data_melanoma_filtered$Year)
  max_year <- max(data_melanoma_filtered$Year)
  year_list <- c(min_year, unname(unlist(joinpoint_years)), max_year)
  segment_years <- matrix(nrow = num_breakpoints + 1, ncol = 2)
  for (i in 2:length(year_list)) {
    segment_years[i - 1, 1] <- year_list[i - 1]
    segment_years[i - 1, 2] <- year_list[i]
  }

  # APC / AAPC
  apc <- slope(out.glm.best, APC = TRUE)$Year
  aapc <- aapc(out.glm.best, parm = "Year")

  return(list(
    race = race,
    sex = sex,
    site = site,
    num_breakpoints = num_breakpoints,
    joinpoint_years = joinpoint_years,
    segment_years = segment_years,
    apc = apc,
    aapc = aapc,
    fit = out.glm.best,
    data = data_melanoma_filtered
```

```
  ))
}
```

```
races <- c("All races", "Black", "White")
sexes <- c("Male and female", "Male", "Female")
sites <- c("All sites", "Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)")

combos_race <- c()
combos_sex <- c()
combos_site <- c()
model_list <- list()

num_breakpoints_list <- c()
joinpoint_years_list <- list()
segment_years_list <- list()
apc_list <- list()
aapc_list <- list()

for (race in races) {
  for (sex in sexes) {
    for (site in sites) {
      model_obj <- joinpoint_model(data_melanoma, race, sex, site)
      if (is.null(model_obj)) next
      combos_race <- c(combos_race, race)
      combos_sex <- c(combos_sex, sex)
      combos_site <- c(combos_site, site)
      model_list[[length(model_list) + 1]] <- model_obj

      num_breakpoints_list <- c(num_breakpoints_list, model_obj$num_breakpoints)
      joinpoint_years_list[[length(joinpoint_years_list) + 1]] <- model_obj$joinpoint_years
      segment_years_list[[length(segment_years_list) + 1]] <- model_obj$segment_years
      apc_list[[length(apc_list) + 1]] <- model_obj$apc
      aapc_list[[length(aapc_list) + 1]] <- model_obj$aapc
    }
  }
}

models <- data.frame(
  Race = combos_race,
  Sex = combos_sex,
  Site = combos_site,
  Num_Breakpoints = num_breakpoints_list,
  Joinpoint_Years = I(joinpoint_years_list),
  Segment_Years = I(segment_years_list),
  APC = I(apc_list),
  AAPC = I(aapc_list),
  Model = I(model_list),
  stringsAsFactors = FALSE
)
```

```
models
```

```
       Race             Sex                     Site Num_Breakpoints
1  All races Male and female                All sites               5
2  All races Male and female            Head and Neck               5
3  All races Male and female                    Trunk               5
4  All races Male and female Arm (shoulder to finger)               4
5  All races Male and female         Leg (hip to toe)               6
6  All races            Male                All sites               5
7  All races            Male            Head and Neck               5
8  All races            Male                    Trunk               5
9  All races            Male Arm (shoulder to finger)               5
10 All races            Male         Leg (hip to toe)               5
11 All races          Female                All sites               4
12 All races          Female            Head and Neck               4
13 All races          Female                    Trunk               4
14 All races          Female Arm (shoulder to finger)               6
15 All races          Female         Leg (hip to toe)               6
16     Black Male and female                All sites               5
17     Black Male and female            Head and Neck               5
18     Black Male and female                    Trunk               1
19     Black Male and female Arm (shoulder to finger)               3
20     Black Male and female         Leg (hip to toe)               5
21     Black            Male                All sites               5
22     Black            Male            Head and Neck               4
23     Black            Male                    Trunk               3
24     Black            Male Arm (shoulder to finger)               1
25     Black            Male         Leg (hip to toe)               5
26     Black          Female                All sites               5
27     Black          Female            Head and Neck               5
28     Black          Female Arm (shoulder to finger)               5
29     Black          Female         Leg (hip to toe)               7
30     White Male and female                All sites               2
31     White Male and female            Head and Neck               5
32     White Male and female                    Trunk               5
33     White Male and female Arm (shoulder to finger)               5
34     White Male and female         Leg (hip to toe)               6
35     White            Male                All sites               6
36     White            Male            Head and Neck               5
37     White            Male                    Trunk               5
38     White            Male Arm (shoulder to finger)               5
39     White            Male         Leg (hip to toe)               6
40     White          Female                All sites               5
41     White          Female            Head and Neck               5
42     White          Female                    Trunk               5
43     White          Female Arm (shoulder to finger)               5
44     White          Female         Leg (hip to toe)               6
   Joinpoint_Years Segment_Years          APC         AAPC        Model
1      1985, 19....  1975, 19.... 4.3892, .... 0.024123.... All race....
2      1977, 19....  1975, 19.... 1.8027, .... 0.024301.... All race....
```

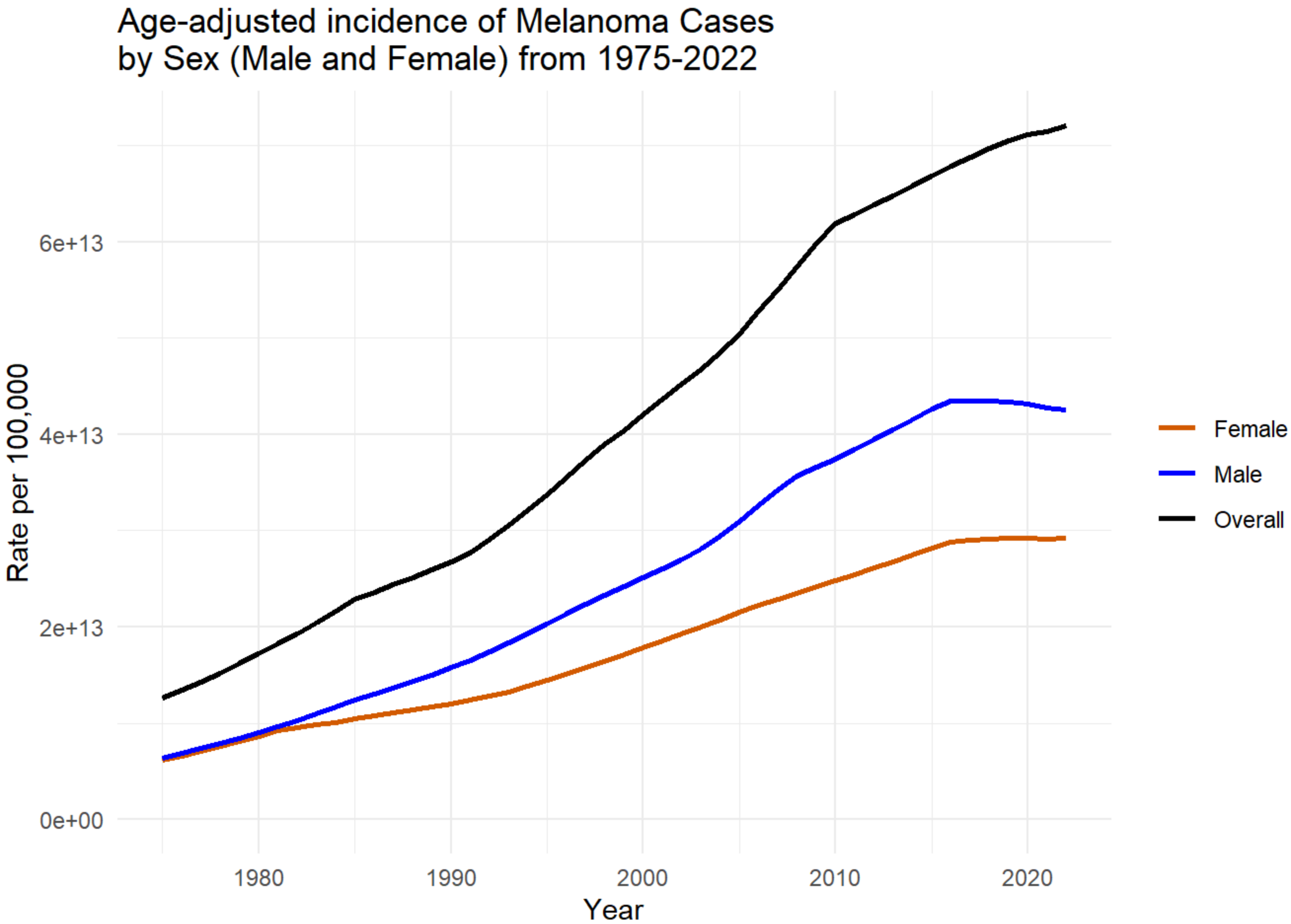
```
3     1976, 19....  1975, 19.... 6.2744, .... 0.024452.... All race....
4     1980, 19....  1975, 19.... 6.0689, .... 0.026738.... All race....
5     1980, 19....  1975, 19.... 1.1386, .... 0.016755.... All race....
6     1985, 19....  1975, 19.... 5.0773, .... 0.026843.... All race....
7     1985, 19....  1975, 19.... 5.2029, .... 0.030102.... All race....
8     1979, 19....  1975, 19.... 7.3627, .... 0.024365.... All race....
9     1987, 19....  1975, 19.... 5.1796, .... 0.030825.... All race....
10    1982, 19....  1975, 19.... 4.7524, .... 0.023697.... All race....
11    1981, 19....  1975, 19.... 4.9915, .... 0.020591.... All race....
12    1981, 19....  1975, 19.... 2.3685, .... 0.016026.... All race....
13    1988, 20....  1975, 19.... 4.2237, .... 0.024070.... All race....
14    1981, 19....  1975, 19.... 6.1517, .... 0.024245.... All race....
15    1982, 19....  1975, 19.... 4.4056, .... 0.016307.... All race....
16    1978, 19....  1975, 19.... -26.174,.... -0.01796.... Black, M....
17    1981, 19....  1975, 19.... -32.085,.... -0.02769.... Black, M....
18            1986  1975, 19.... 9.5118, .... 0.022126.... Black, M....
19    1986, 19....  1975, 19.... 3.6818, .... 0.007409.... Black, M....
20    1977, 19....  1975, 19.... -23.567,.... -0.01212.... Black, M....
21    1978, 19....  1975, 19.... -40.215,.... -0.03001.... Black, M....
22    1980, 19....  1975, 19.... -39.442,.... -0.00376.... Black, M....
23    1987, 20....  1975, 19.... 43.326, .... 0.057983.... Black, M....
24            1989  1975, 19.... 19.793, .... 0.050850.... Black, M....
25    1979, 19....  1975, 19.... -37.182,.... -0.04622.... Black, M....
26    1978, 19....  1975, 19.... -11.836,.... -0.00461.... Black, F....
27    1982, 19....  1975, 19.... -31.715,.... -0.04061.... Black, F....
28    1981, 19....  1975, 19.... -15.815,.... -0.01437.... Black, F....
29    1978, 19....  1975, 19.... 13.542, .... 0.008872.... Black, F....
30      1981, 2009  1975, 19.... 5.6119, .... 0.028216.... White, M....
31    1982, 19....  1975, 19.... 3.5851, .... 0.028418.... White, M....
32    1980, 19....  1975, 19.... 7.5944, .... 0.029151.... White, M....
33    1980, 19....  1975, 19.... 6.3003, .... 0.030143.... White, M....
34    1980, 19....  1975, 19.... 2.0633, .... 0.020761.... White, M....
35    1983, 19....  1975, 19.... 5.7594, .... 0.030943.... White, M....
36    1994, 20....  1975, 19.... 5.1138, .... 0.033626.... White, M....
37    1986, 19....  1975, 19.... 5.8485, .... 0.026774.... White, M....
38    1980, 19....  1975, 19.... 6.1715, .... 0.034051.... White, M....
39    1989, 20....  1975, 19.... 3.0322, .... 0.025072.... White, M....
40    1982, 19....  1975, 19.... 4.9671, .... 0.024623.... White, F....
41    1980, 19....  1975, 19.... 3.4722, .... 0.020628.... White, F....
42    1978, 19....  1975, 19.... 11.423, .... 0.031297.... White, F....
43    1981, 19....  1975, 19.... 6.1728, .... 0.027168.... White, F....
44    1977, 19....  1975, 19.... 1.9372, .... 0.019701.... White, F....
```

```r
get_model <- function(models, race, sex, site) {
  idx <- which(models$Race == race
             & models$Sex == sex
             & models$Site == site)
  if (length(idx) > 1) {
    idx <- idx[1]
  }
```

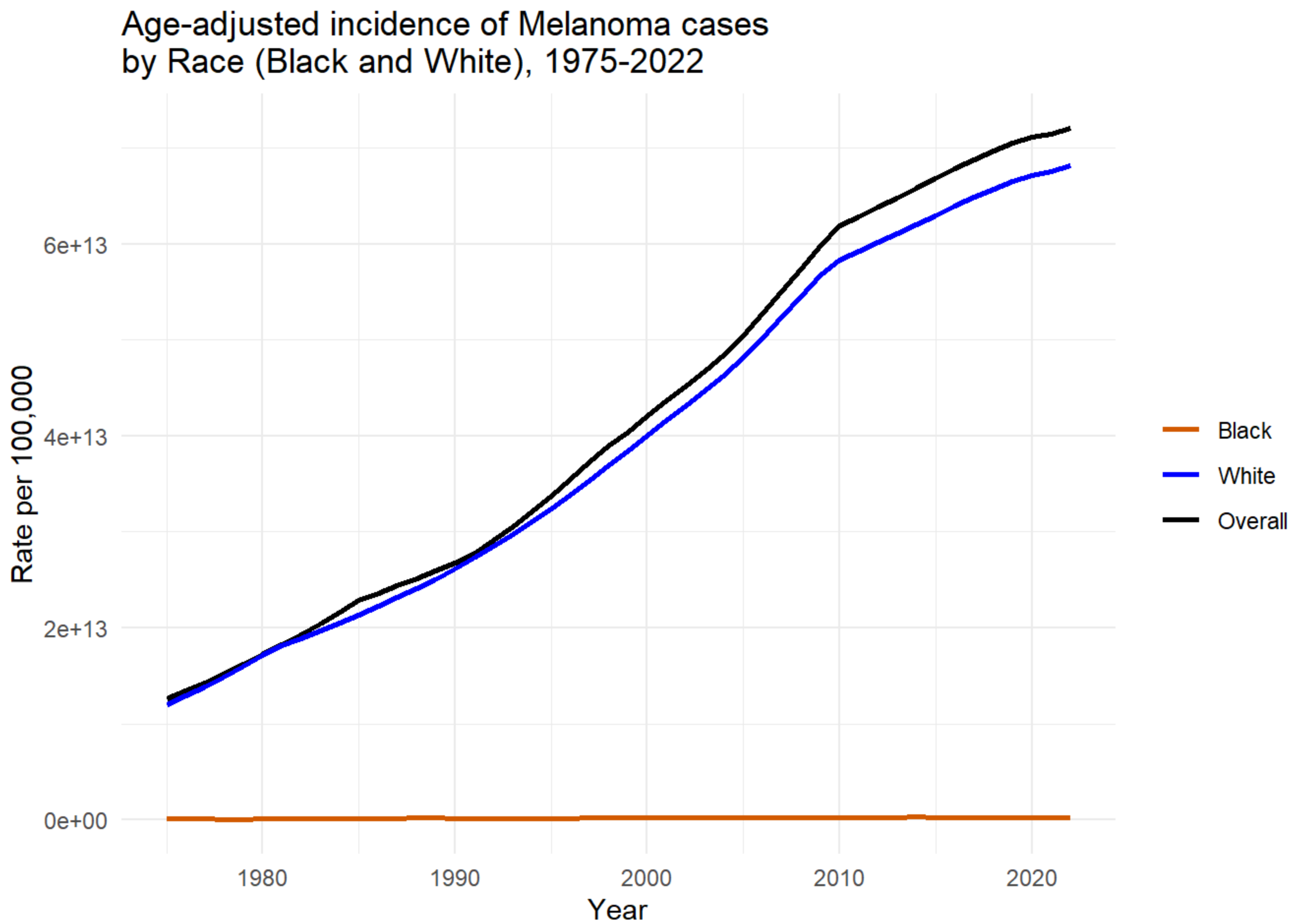
```
  models$Model[[idx]]
}
```

```
plot_joinpoint_set <- function(model_objs, labels, cols,
                               main, ylab = "Rate per 100,000") {
  df_list <- list()
  for (i in seq_along(model_objs)) {
    m <- model_objs[[i]]
    d <- m$data

    year_grid <- seq(min(d$Year), max(d$Year), by = 1)
    pred_rate <- exp(predict(m$fit, newdata = data.frame(Year = year_grid),
                             type = "link")) * 100000

    df_list[[i]] <- rbind(
      data.frame(Year = d$Year, Rate = d$Rate, Series = labels[i], Type = "Observed"),
      data.frame(Year = year_grid, Rate = pred_rate, Series = labels[i], Type = "Fitted")
    )
  }
  plot_df <- do.call(rbind, df_list)

  ylim_max <- max(plot_df$Rate[plot_df$Type == "Fitted"], na.rm = TRUE)

  ggplot(plot_df, aes(x = Year, y = Rate, group = Series)) +
    geom_line(data = subset(plot_df, Type == "Fitted"), aes(color = Series), linewidth = 1) +
    scale_color_manual(values = cols, breaks = labels) +
    coord_cartesian(ylim = c(0, ylim_max)) +
    labs(title = main, y = ylab, color = NULL) +
    theme_minimal()
}
```

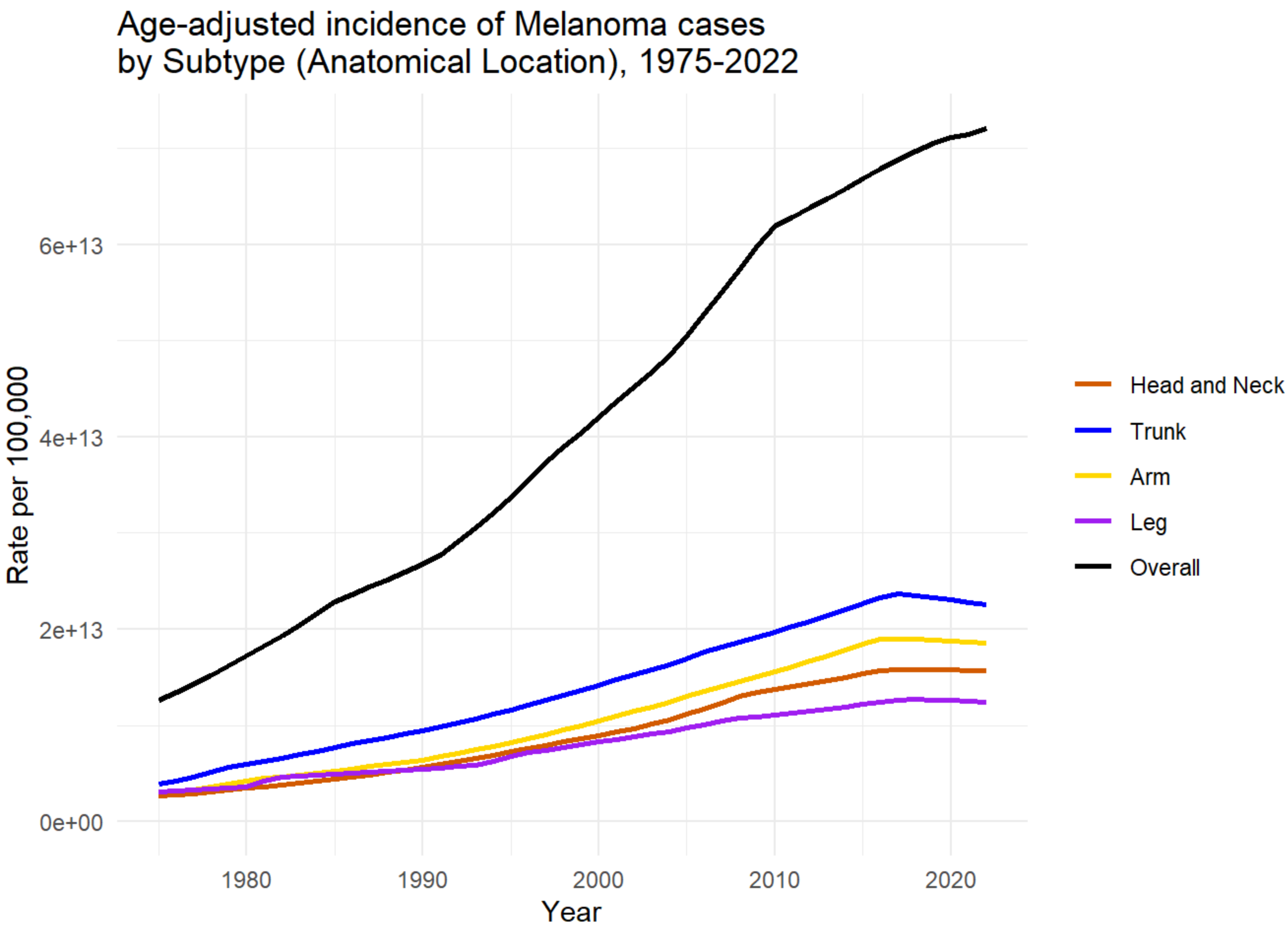
```
# Stratified by Sex
m_overall <- get_model(models, "All races", "Male and female", "All sites")
m_male    <- get_model(models, "All races", "Male", "All sites")
m_female  <- get_model(models, "All races", "Female", "All sites")

plot_joinpoint_set(
  list(m_female, m_male, m_overall),
  labels = c("Female", "Male", "Overall"),
  cols   = c("#D55E00", "blue", "black"),
  main   = "Age-adjusted incidence of Melanoma Cases\nby Sex (Male and Female) from 1975-2022"
)
```



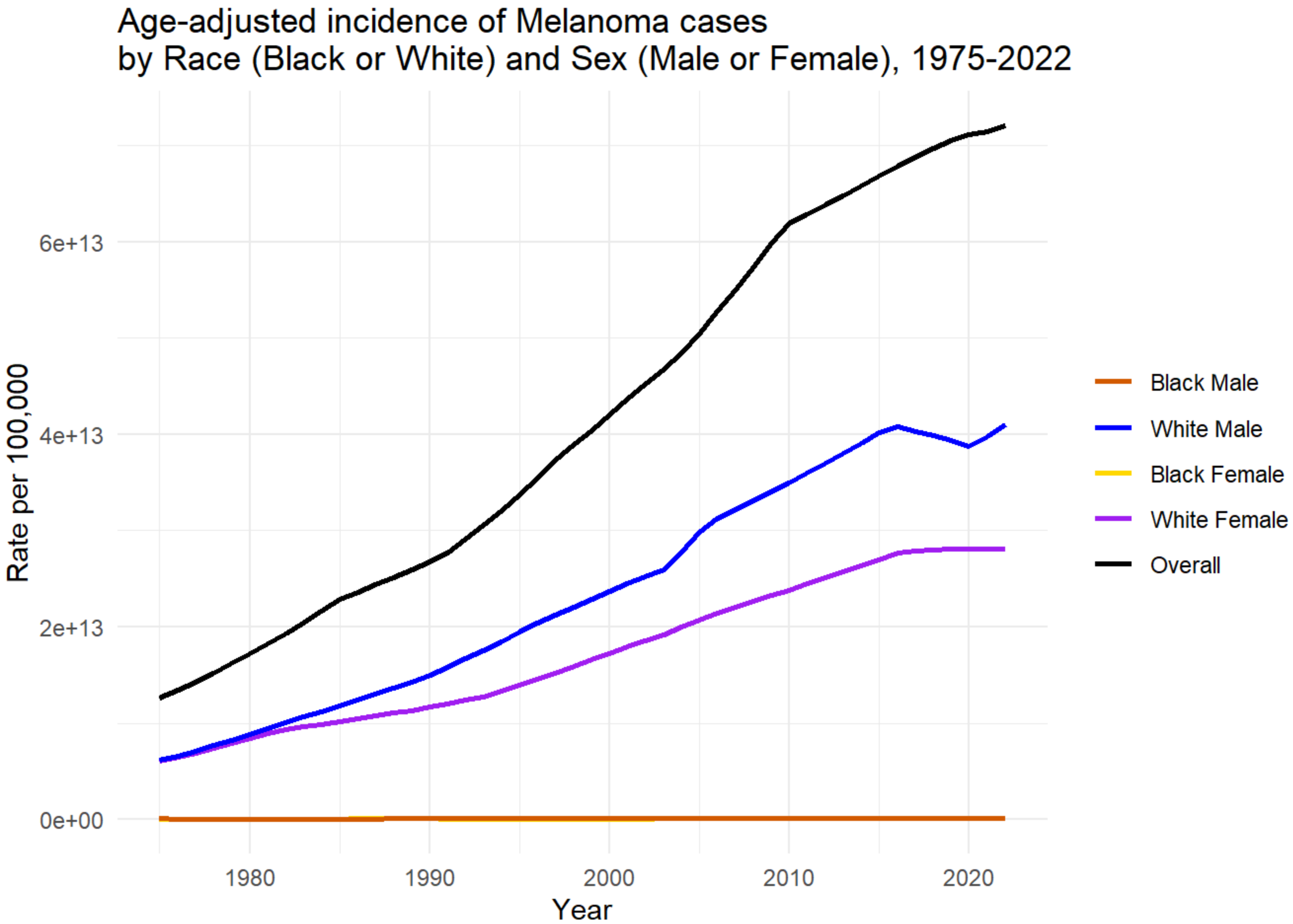

```
# Stratified by Race
m_black <- get_model(models, "Black", "Male and female", "All sites")
m_white <- get_model(models, "White", "Male and female", "All sites")

plot_joinpoint_set(
  list(m_black, m_white, m_overall),
  labels = c("Black", "White", "Overall"),
  cols   = c("#D55E00", "blue", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Race (Black and White), 1975-2022"
)
```



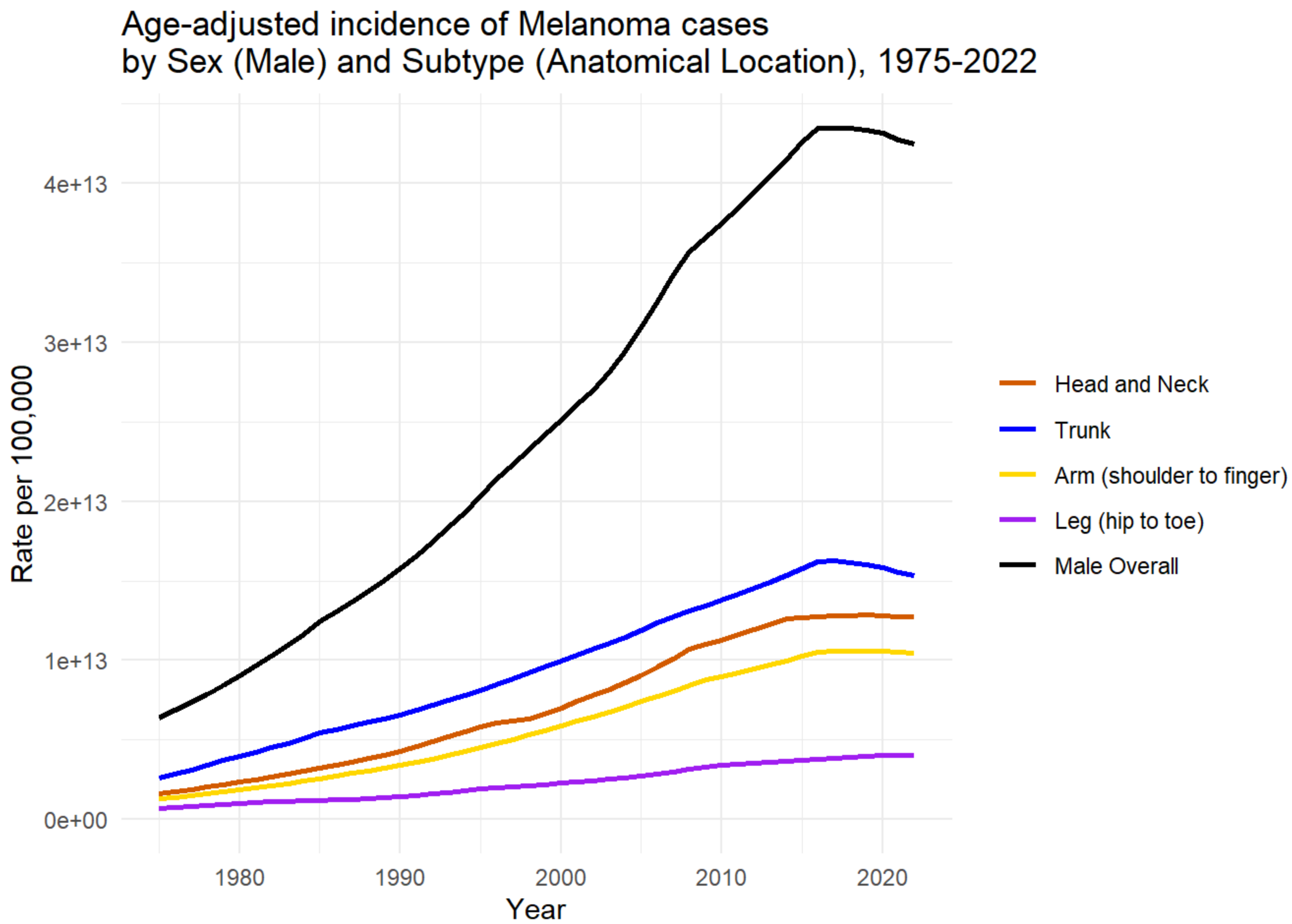

```
# Stratified by anatomical location (Site)
m_headneck <- get_model(models, "All races", "Male and female", "Head and Neck")
m_trunk    <- get_model(models, "All races", "Male and female", "Trunk")
m_arm      <- get_model(models, "All races", "Male and female", "Arm (shoulder to finger)")
m_leg      <- get_model(models, "All races", "Male and female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_headneck, m_trunk, m_arm, m_leg, m_overall),
  labels = c("Head and Neck", "Trunk", "Arm", "Leg", "Overall"),
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Subtype (Anatomical Location), 1975-2022
)
```



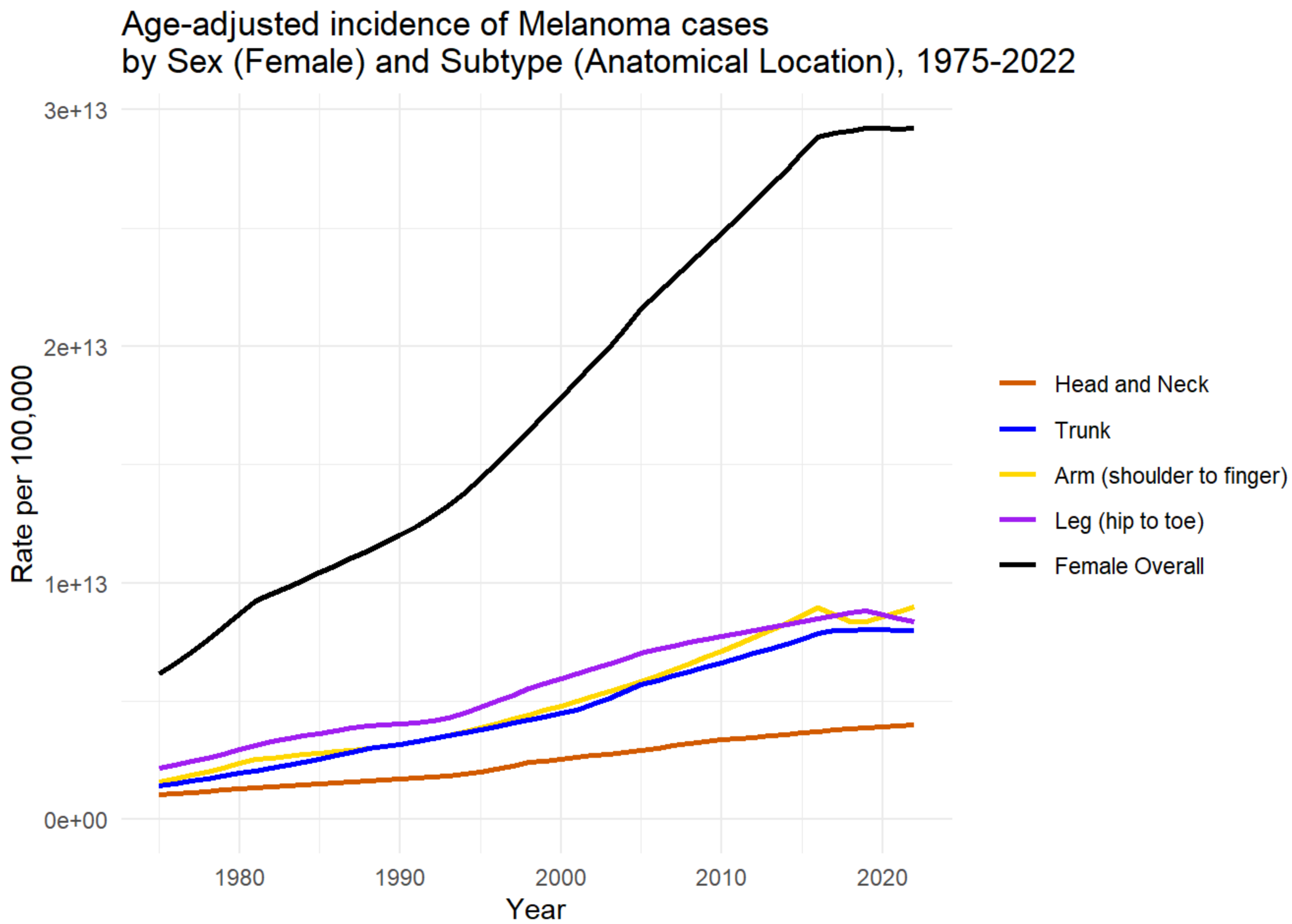

```r
# Stratified by Race (Black/White) and Sex (Male/Female)
m_black_male    <- get_model(models, "Black", "Male", "All sites")
m_white_male    <- get_model(models, "White", "Male", "All sites")
m_black_female  <- get_model(models, "Black", "Female", "All sites")
m_white_female  <- get_model(models, "White", "Female", "All sites")

plot_joinpoint_set(
  list(m_black_male, m_white_male, m_black_female, m_white_female, m_overall),
  labels = c("Black Male", "White Male", "Black Female", "White Female", "Overall"),
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Race (Black or White) and Sex (Male or Fe
)
```



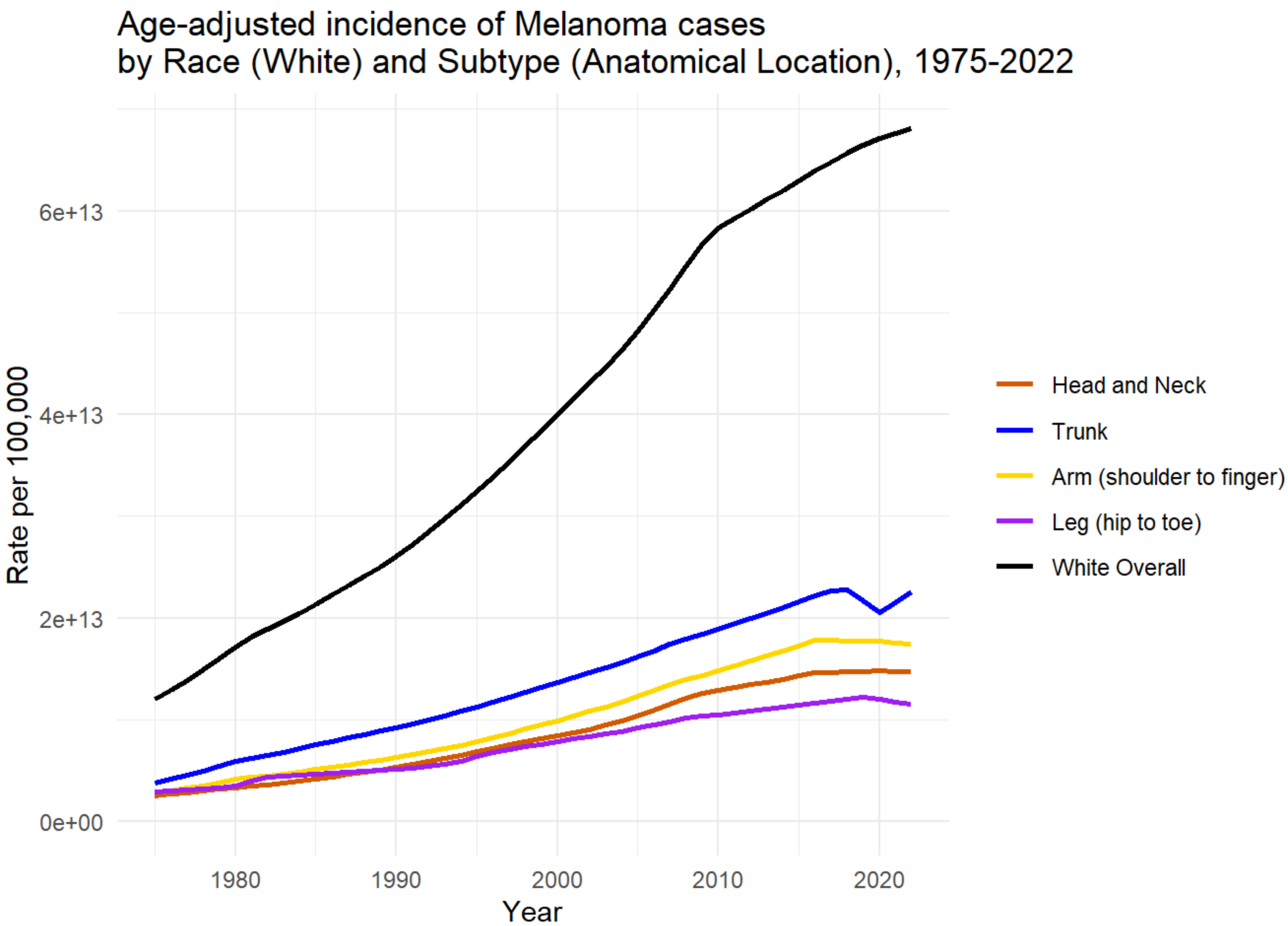

```r
# Stratified by Sex (Male) and Site
m_male_headneck <- get_model(models, "All races", "Male", "Head and Neck")
m_male_trunk    <- get_model(models, "All races", "Male", "Trunk")
m_male_arm      <- get_model(models, "All races", "Male", "Arm (shoulder to finger)")
m_male_leg      <- get_model(models, "All races", "Male", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_male_headneck, m_male_trunk, m_male_arm, m_male_leg, m_male),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "Male Overa
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Sex (Male) and Subtype (Anatomical Locat
)
```



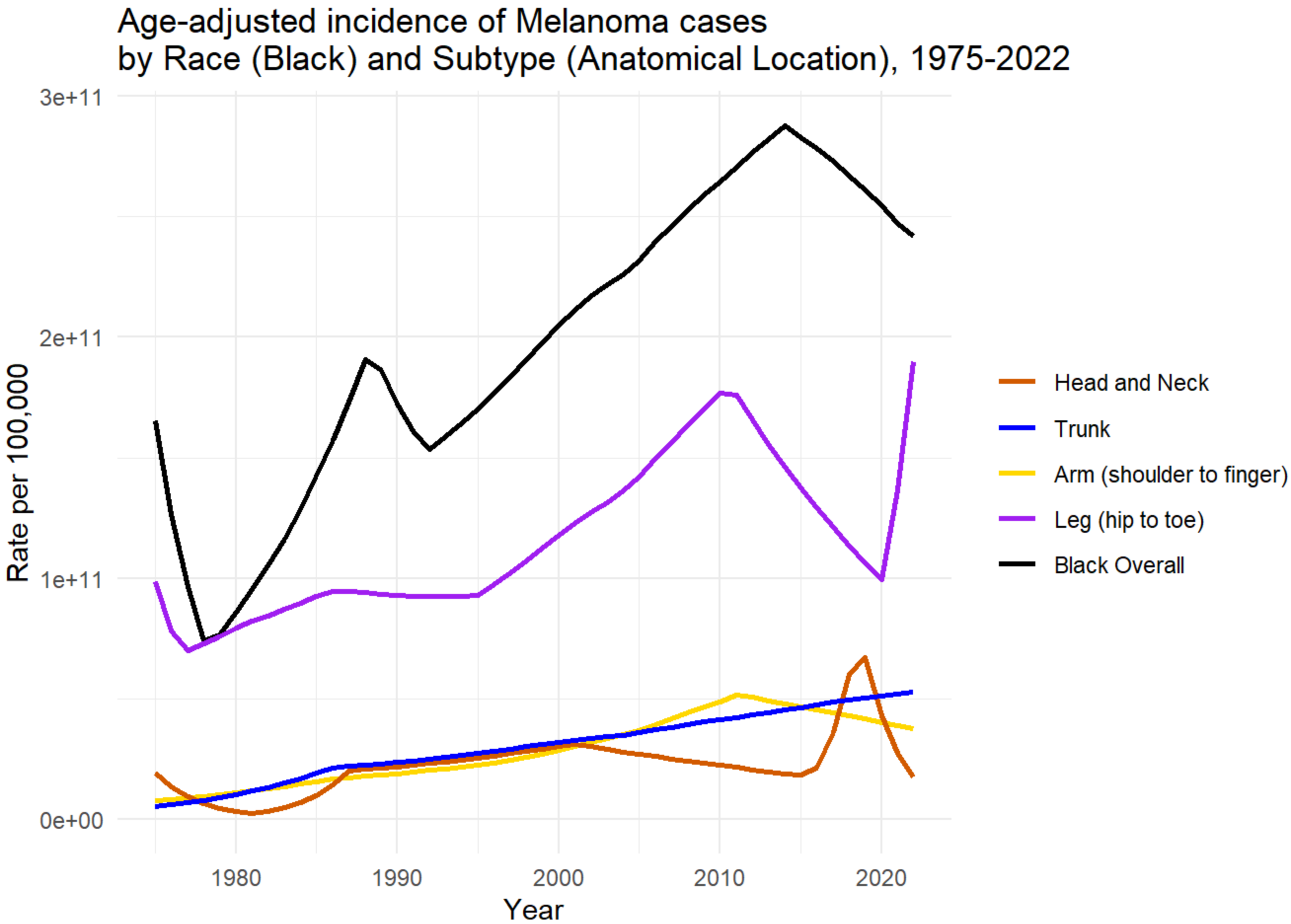

```
# Stratified by Sex (Female) and Site
m_female_headneck <- get_model(models, "All races", "Female", "Head and Neck")
m_female_trunk    <- get_model(models, "All races", "Female", "Trunk")
m_female_arm      <- get_model(models, "All races", "Female", "Arm (shoulder to finger)")
m_female_leg      <- get_model(models, "All races", "Female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_female_headneck, m_female_trunk, m_female_arm, m_female_leg, m_female),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "Female Ov
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Sex (Female) and Subtype (Anatomical Loc
)
```




```
# Stratified by Race (White) and Site
m_white_headneck <- get_model(models, "White", "Male and female", "Head and Neck")
m_white_trunk    <- get_model(models, "White", "Male and female", "Trunk")
m_white_arm      <- get_model(models, "White", "Male and female", "Arm (shoulder to finger)")
m_white_leg      <- get_model(models, "White", "Male and female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_white_headneck, m_white_trunk, m_white_arm, m_white_leg,
       get_model(models, "White", "Male and female", "All sites")),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "White Over
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Race (White) and Subtype (Anatomical Loca
)
```




```
# Stratified by Race (Black) and Site
m_black_headneck <- get_model(models, "Black", "Male and female", "Head and Neck")
m_black_trunk    <- get_model(models, "Black", "Male and female", "Trunk")
m_black_arm      <- get_model(models, "Black", "Male and female", "Arm (shoulder to finger)")
m_black_leg      <- get_model(models, "Black", "Male and female", "Leg (hip to toe)")

plot_joinpoint_set(
  list(m_black_headneck, m_black_trunk, m_black_arm, m_black_leg,
       get_model(models, "Black", "Male and female", "All sites")),
  labels = c("Head and Neck", "Trunk", "Arm (shoulder to finger)", "Leg (hip to toe)", "Black Over
  cols   = c("#D55E00", "blue", "gold", "purple", "black"),
  main   = "Age-adjusted incidence of Melanoma cases\nby Race (Black) and Subtype (Anatomical Loca
)
```



Appendix: All Model Output

```
for (i in seq_len(nrow(models))) {
  cat("\n======================\n")
  cat("Row:", i, "\n")
  cat("Race:", models$Race[i], "\n")
  cat("Sex:", models$Sex[i], "\n")
  cat("Site:", models$Site[i], "\n")
  cat("Num_Breakpoints:", models$Num_Breakpoints[i], "\n")
  cat("Joinpoint_Years:", paste(models$Joinpoint_Years[[i]], collapse = ", "), "\n")
  cat("Segment_Years:\n")
  print(models$Segment_Years[[i]])
  cat("APC:\n")
  print(models$APC[[i]])
  cat("AAPC:\n")
  print(models$AAPC[[i]])
}
```

```
======================
Row: 1
Race: All races
```

```
Sex: Male and female
Site: All sites
Num_Breakpoints: 5
Joinpoint_Years: 1985, 1991, 1997, 2003, 2010
Segment_Years:
     [,1] [,2]
[1,] 1975 1985
[2,] 1985 1991
[3,] 1991 1997
[4,] 1997 2003
[5,] 2003 2010
[6,] 2010 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1 4.38920   4.38740   4.39090
slope2 1.98860   1.98620   1.99100
slope3 3.44880   3.44620   3.45150
slope4 2.58460   2.58220   2.58690
slope5 2.95300   2.95090   2.95510
slope6 0.30478   0.30422   0.30534
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.412362e-02 1.126549e-06 2.412141e-02 2.412583e-02
attr(,"wrong.se")
[1] 3.671217e-06

=======================
Row: 2
Race: All races
Sex: Male and female
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1977, 1996, 2002, 2008, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1977
[2,] 1977 1996
[3,] 1996 2002
[4,] 2002 2008
[5,] 2008 2016
[6,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  1.80270   1.74930   1.85620
slope2  3.61160   3.61040   3.61290
slope3  2.70660   2.70240   2.71070
slope4  3.81940   3.81490   3.82400
slope5  1.01670   1.01420   1.01930
slope6 -0.73923  -0.74293  -0.73552
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
```

```
2.430109e-02 4.274866e-06 2.429271e-02 2.430947e-02
attr(,"wrong.se")
[1] 1.257075e-05

=======================
Row: 3
Race: All races
Sex: Male and female
Site: Trunk
Num_Breakpoints: 5
Joinpoint_Years: 1976, 1979, 1986, 2006, 2017
Segment_Years:
     [,1] [,2]
[1,] 1975 1976
[2,] 1976 1979
[3,] 1979 1986
[4,] 1986 2006
[5,] 2006 2017
[6,] 2017 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  6.2744    6.2276    6.3212
slope2  8.4352    8.3922    8.4781
slope3  3.9148    3.9101    3.9196
slope4  2.5790    2.5783    2.5796
slope5  1.5277    1.5261    1.5292
slope6 -1.5567   -1.5598   -1.5537
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.445228e-02 3.545548e-06 2.444533e-02 2.445922e-02
attr(,"wrong.se")
[1] 1.352031e-05

=======================
Row: 4
Race: All races
Sex: Male and female
Site: Arm (shoulder to finger)
Num_Breakpoints: 4
Joinpoint_Years: 1980, 1990, 2005, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1990
[3,] 1990 2005
[4,] 2005 2016
[5,] 2016 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  6.0689    6.0605    6.0773
slope2  2.7441    2.7411    2.7471
```

```
slope3  3.4477    3.4464    3.4489
slope4  2.1793    2.1775    2.1810
slope5 -1.0904   -1.0931   -1.0877
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.673811e-02 2.967182e-06 2.673230e-02 2.674393e-02
attr(,"wrong.se")
[1] 6.53466e-06

=======================
Row: 5
Race: All races
Sex: Male and female
Site: Leg (hip to toe)
Num_Breakpoints: 6
Joinpoint_Years: 1980, 1981, 1994, 1995, 2008, 2018
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1981
[3,] 1981 1994
[4,] 1994 1995
[5,] 1995 2008
[6,] 2008 2018
[7,] 2018 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  1.13860   1.12770   1.14960
slope2 14.81500  14.76400  14.86600
slope3  0.89296   0.89067   0.89525
slope4  7.89470   7.85770   7.93180
slope5  2.21940   2.21760   2.22120
slope6  0.57039   0.56838   0.57241
slope7 -1.06750  -1.07300  -1.06200
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
1.675554e-02 3.291912e-06 1.674909e-02 1.676199e-02
attr(,"wrong.se")
[1] 1.246792e-05

=======================
Row: 6
Race: All races
Sex: Male
Site: All sites
Num_Breakpoints: 5
Joinpoint_Years: 1985, 1996, 2002, 2008, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1985
[2,] 1985 1996
```

```
[3,] 1996 2002
[4,] 2002 2008
[5,] 2008 2016
[6,] 2016 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  5.0773    5.0752    5.0794
slope2  3.5416    3.5401    3.5431
slope3  2.4795    2.4765    2.4826
slope4  3.8057    3.8020    3.8093
slope5  1.2331    1.2315    1.2346
slope6 -1.1051   -1.1068   -1.1033
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.684379e-02 1.555562e-06 2.684074e-02 2.684684e-02
attr(,"wrong.se")
[1] 4.345341e-06

=======================
Row: 7
Race: All races
Sex: Male
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1985, 1995, 1998, 2008, 2014
Segment_Years:
     [,1] [,2]
[1,] 1975 1985
[2,] 1985 1995
[3,] 1995 1998
[4,] 1998 2008
[5,] 2008 2014
[6,] 2014 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  5.20290   5.19800   5.20770
slope2  4.60830   4.60530   4.61130
slope3  0.24773   0.21231   0.28316
slope4  4.19290   4.19070   4.19500
slope5  1.39730   1.39300   1.40170
slope6 -0.77973  -0.78239  -0.77707
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.010208e-02 3.102512e-06 3.009600e-02 3.010816e-02
attr(,"wrong.se")
[1] 1.233178e-05

=======================
Row: 8
Race: All races
Sex: Male
```

```
Site: Trunk
Num_Breakpoints: 5
Joinpoint_Years: 1979, 1985, 1995, 2006, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1979
[2,] 1979 1985
[3,] 1985 1995
[4,] 1995 2006
[5,] 2006 2016
[6,] 2016 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  7.3627    7.3453    7.3801
slope2  4.9682    4.9624    4.9740
slope3  2.6280    2.6253    2.6307
slope4  2.5845    2.5826    2.5864
slope5  1.3455    1.3437    1.3473
slope6 -1.7853   -1.7890   -1.7817
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.436597e-02 3.611422e-06 2.435889e-02 2.437305e-02
attr(,"wrong.se")
[1] 8.885244e-06

=======================
Row: 9
Race: All races
Sex: Male
Site: Arm (shoulder to finger)
Num_Breakpoints: 5
Joinpoint_Years: 1987, 1996, 2004, 2009, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1987
[2,] 1987 1996
[3,] 1996 2004
[4,] 2004 2009
[5,] 2009 2016
[6,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  5.17960   5.17610   5.18300
slope2  4.09680   4.09160   4.10190
slope3  3.80020   3.79670   3.80360
slope4  2.86720   2.86020   2.87430
slope5  1.29900   1.29520   1.30290
slope6 -0.86421  -0.86874  -0.85969
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.082575e-02 3.179979e-06 3.081952e-02 3.083199e-02
```

```
attr(,"wrong.se")
[1] 8.89905e-06

=======================
Row: 10
Race: All races
Sex: Male
Site: Leg (hip to toe)
Num_Breakpoints: 5
Joinpoint_Years: 1982, 1988, 1995, 2003, 2010
Segment_Years:
     [,1] [,2]
[1,] 1975 1982
[2,] 1982 1988
[3,] 1988 1995
[4,] 1995 2003
[5,] 2003 2010
[6,] 2010 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1 4.75240   4.73910   4.76570
slope2 1.14080   1.13000   1.15160
slope3 4.22510   4.21550   4.23470
slope4 2.17940   2.17130   2.18750
slope5 3.22430   3.21720   3.23140
slope6 0.37036   0.36797   0.37275
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.369793e-02 5.491543e-06 2.368716e-02 2.370869e-02
attr(,"wrong.se")
[1] 1.651582e-05

=======================
Row: 11
Race: All races
Sex: Female
Site: All sites
Num_Breakpoints: 4
Joinpoint_Years: 1981, 1993, 2005, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 1993
[3,] 1993 2005
[4,] 2005 2016
[5,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  4.99150    4.9870   4.99600
slope2  1.69660    1.6948   1.69840
slope3  2.86930    2.8681   2.87040
```

```
slope4  1.46020    1.4588   1.46160
slope5 -0.43473   -0.4369  -0.43257
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.059127e-02 1.967540e-06 2.058741e-02 2.059513e-02
attr(,"wrong.se")
[1] 4.467012e-06

=======================
Row: 12
Race: All races
Sex: Female
Site: Head and Neck
Num_Breakpoints: 4
Joinpoint_Years: 1981, 1994, 1998, 2009
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 1994
[3,] 1994 1998
[4,] 1998 2009
[5,] 2009 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1 2.36850   2.35440   2.38250
slope2 1.38160   1.37830   1.38490
slope3 4.76060   4.72930   4.79210
slope4 1.70760   1.70450   1.71080
slope5 0.55665   0.55424   0.55906
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
1.602675e-02 5.007520e-06 1.601693e-02 1.603656e-02
attr(,"wrong.se")
[1] 1.639533e-05

=======================
Row: 13
Race: All races
Sex: Female
Site: Trunk
Num_Breakpoints: 4
Joinpoint_Years: 1988, 2001, 2005, 2017
Segment_Years:
     [,1] [,2]
[1,] 1975 1988
[2,] 1988 2001
[3,] 2001 2005
[4,] 2005 2017
[5,] 2017 2022
APC:
           Est. CI(95%).l CI(95%).u
```

```
slope1  4.22370    4.2203   4.22700
slope2  1.99560    1.9931   1.99800
slope3  4.39300    4.3800   4.40600
slope4  1.70380    1.7017   1.70580
slope5 -0.50278   -0.5080  -0.49757
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.407012e-02 3.211613e-06 2.406383e-02 2.407641e-02
attr(,"wrong.se")
[1] 8.817169e-06

=======================
Row: 14
Race: All races
Sex: Female
Site: Arm (shoulder to finger)
Num_Breakpoints: 6
Joinpoint_Years: 1981, 1989, 1993, 2005, 2016, 2018
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 1989
[3,] 1989 1993
[4,] 1993 2005
[5,] 2005 2016
[6,] 2016 2018
[7,] 2018 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  6.1517    6.1404    6.1629
slope2  1.1531    1.1482    1.1579
slope3  2.1274    2.1029    2.1518
slope4  2.9705    2.9680    2.9730
slope5  2.7074    2.7055    2.7094
slope6 -4.0800   -4.1088   -4.0512
slope7  1.8504    1.8408    1.8600
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.424533e-02 4.166088e-06 2.423716e-02 2.425349e-02
attr(,"wrong.se")
[1] 1.580302e-05

=======================
Row: 15
Race: All races
Sex: Female
Site: Leg (hip to toe)
Num_Breakpoints: 6
Joinpoint_Years: 1982, 1987, 1993, 1998, 2005, 2019
Segment_Years:
     [,1] [,2]
```

```
[1,] 1975 1982
[2,] 1982 1987
[3,] 1987 1993
[4,] 1993 1998
[5,] 1998 2005
[6,] 2005 2019
[7,] 2019 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  4.40560   4.39790   4.41320
slope2  2.05190   2.04130   2.06240
slope3  0.13172   0.12432   0.13912
slope4  3.68020   3.67320   3.68720
slope5  2.36300   2.35820   2.36780
slope6  0.45122   0.44976   0.45268
slope7 -2.21460  -2.22930  -2.19980
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
1.630779e-02 3.661934e-06 1.630061e-02 1.631497e-02
attr(,"wrong.se")
[1] 1.191201e-05

=======================
Row: 16
Race: Black
Sex: Male and female
Site: All sites
Num_Breakpoints: 5
Joinpoint_Years: 1978, 1988, 1992, 2003, 2014
Segment_Years:
     [,1] [,2]
[1,] 1975 1978
[2,] 1978 1988
[3,] 1988 1992
[4,] 1992 2003
[5,] 2003 2014
[6,] 2014 2022
APC:
             Est.  CI(95%).l  CI(95%).u
slope1 -26.174000 -26.237000 -26.110000
slope2   7.329200   7.308100   7.350300
slope3  -9.517700  -9.605300  -9.430000
slope4   0.663010   0.648720   0.677310
slope5  -0.037306  -0.049133  -0.025478
slope6  -3.906900  -3.921800  -3.892000
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-1.796532e-02  1.666616e-05 -1.799798e-02 -1.793265e-02
attr(,"wrong.se")
[1] 5.75047e-05
```

```
======================
Row: 17
Race: Black
Sex: Male and female
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1981, 1987, 2001, 2016, 2019
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 1987
[3,] 1987 2001
[4,] 2001 2016
[5,] 2016 2019
[6,] 2019 2022
APC:
            Est. CI(95%).l CI(95%).u
slope1 -32.08500 -32.20200 -31.96700
slope2  37.68400  37.49300  37.87500
slope3   0.38465   0.35629   0.41302
slope4  -6.09110  -6.11590  -6.06630
slope5  64.95000  64.55200  65.34900
slope6 -36.72300 -36.81900 -36.62700
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-0.0276983320  0.0000543063 -0.0278047704 -0.0275918936
attr(,"wrong.se")
[1] 0.000182854

======================
Row: 18
Race: Black
Sex: Male and female
Site: Trunk
Num_Breakpoints: 1
Joinpoint_Years: 1986
Segment_Years:
     [,1] [,2]
[1,] 1975 1986
[2,] 1986 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1 9.51180   9.43840   9.58530
slope2 0.12819   0.12312   0.13327
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.212617e-02 4.022642e-05 2.204733e-02 2.220501e-02
attr(,"wrong.se")
[1] 8.207011e-05

======================
```

```
Row: 19
Race: Black
Sex: Male and female
Site: Arm (shoulder to finger)
Num_Breakpoints: 3
Joinpoint_Years: 1986, 1998, 2011
Segment_Years:
     [,1] [,2]
[1,] 1975 1986
[2,] 1986 1998
[3,] 1998 2011
[4,] 2011 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  3.68180   3.62760   3.73600
slope2  0.69227   0.65658   0.72798
slope3  2.93610   2.90990   2.96230
slope4 -4.75020  -4.77800  -4.72230
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
0.0074091756 0.0000404334 0.0073299276 0.0074884236
attr(,"wrong.se")
[1] 9.253994e-05

=======================
Row: 20
Race: Black
Sex: Male and female
Site: Leg (hip to toe)
Num_Breakpoints: 5
Joinpoint_Years: 1977, 1986, 1995, 2011, 2020
Segment_Years:
     [,1] [,2]
[1,] 1975 1977
[2,] 1977 1986
[3,] 1986 1995
[4,] 1995 2011
[5,] 2011 2020
[6,] 2020 2022
APC:
            Est.  CI(95%).l CI(95%).u
slope1 -23.56700 -23.780000 -23.35500
slope2   0.12795   0.098438   0.15748
slope3  -2.80420  -2.829500  -2.77890
slope4   1.68210   1.672600   1.69160
slope5  -8.04110  -8.061100  -8.02110
slope6  36.03800  35.879000  36.19800
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-1.212609e-02  2.835598e-05 -1.218167e-02 -1.207052e-02
attr(,"wrong.se")
```

```
[1] 7.122944e-05

=======================
Row: 21
Race: Black
Sex: Male
Site: All sites
Num_Breakpoints: 5
Joinpoint_Years: 1978, 1989, 1995, 2005, 2019
Segment_Years:
     [,1] [,2]
[1,] 1975 1978
[2,] 1978 1989
[3,] 1989 1995
[4,] 1995 2005
[5,] 2005 2019
[6,] 2019 2022
APC:
            Est. CI(95%).l CI(95%).u
slope1 -40.21500 -40.28400 -40.14500
slope2   9.11960   9.09150   9.14770
slope3   3.92330   3.84210   4.00460
slope4  -3.18180  -3.19880  -3.16480
slope5  -0.36438  -0.37503  -0.35374
slope6 -16.10800 -16.17100 -16.04400
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-3.001581e-02  2.410735e-05 -3.006306e-02 -2.996856e-02
attr(,"wrong.se")
[1] 7.894471e-05

=======================
Row: 22
Race: Black
Sex: Male
Site: Head and Neck
Num_Breakpoints: 4
Joinpoint_Years: 1980, 1995, 2005, 2014
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1995
[3,] 1995 2005
[4,] 2005 2014
[5,] 2014 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1 -39.4420  -39.5710  -39.3130
slope2  23.4600   23.3960   23.5240
slope3 -12.7020  -12.7420  -12.6620
slope4  -2.2509   -2.3088   -2.1929
```

```
slope5  13.4050   13.3400   13.4710
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-3.768170e-03  5.348613e-05 -3.873001e-03 -3.663339e-03
attr(,"wrong.se")
[1] 0.0001748859

=======================
Row: 23
Race: Black
Sex: Male
Site: Trunk
Num_Breakpoints: 3
Joinpoint_Years: 1987, 2002, 2017
Segment_Years:
     [,1] [,2]
[1,] 1975 1987
[2,] 1987 2002
[3,] 2002 2017
[4,] 2017 2022
APC:
            Est. CI(95%).l CI(95%).u
slope1  43.32600  43.17000  43.48100
slope2  -4.40720  -4.43020  -4.38410
slope3   0.73207   0.71099   0.75315
slope4 -15.79100 -15.92000 -15.66300
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
0.0579839514 0.0001163175 0.0577559733 0.0582119295
attr(,"wrong.se")
[1] 0.0001675746

=======================
Row: 24
Race: Black
Sex: Male
Site: Arm (shoulder to finger)
Num_Breakpoints: 1
Joinpoint_Years: 1989
Segment_Years:
     [,1] [,2]
[1,] 1975 1989
[2,] 1989 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1 19.79300   19.7050  19.88200
slope2 -0.41851   -0.4258  -0.41121
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
5.085023e-02 7.964596e-05 5.069412e-02 5.100633e-02
attr(,"wrong.se")
```

```
[1] 0.0001154459

=======================
Row: 25
Race: Black
Sex: Male
Site: Leg (hip to toe)
Num_Breakpoints: 5
Joinpoint_Years: 1979, 1990, 1992, 1999, 2015
Segment_Years:
     [,1] [,2]
[1,] 1975 1979
[2,] 1979 1990
[3,] 1990 1992
[4,] 1992 1999
[5,] 1999 2015
[6,] 2015 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1 -37.1820  -37.2440  -37.1200
slope2  11.7670   11.7270   11.8070
slope3 -60.3690  -60.4630  -60.2750
slope4  25.9010   25.8260   25.9770
slope5  -1.7457   -1.7567   -1.7348
slope6 -13.5080  -13.5520  -13.4650
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-4.622662e-02  2.759254e-05 -4.628070e-02 -4.617254e-02
attr(,"wrong.se")
[1] 0.0001008995

=======================
Row: 26
Race: Black
Sex: Female
Site: All sites
Num_Breakpoints: 5
Joinpoint_Years: 1978, 1987, 2002, 2005, 2020
Segment_Years:
     [,1] [,2]
[1,] 1975 1978
[2,] 1978 1987
[3,] 1987 2002
[4,] 2002 2005
[5,] 2005 2020
[6,] 2020 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1 -11.8360  -12.0050  -11.6680
slope2   3.7377    3.7039    3.7714
slope3  -4.7563   -4.7693   -4.7433
```

```
slope4  21.4610   21.3490   21.5720
slope5  -3.8725   -3.8837   -3.8614
slope6  14.4700   14.3320   14.6080
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-0.0046168707  0.0000303561 -0.0046763676 -0.0045573739
attr(,"wrong.se")
[1] 8.541886e-05

=======================
Row: 27
Race: Black
Sex: Female
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1982, 1984, 1991, 2016, 2018
Segment_Years:
     [,1] [,2]
[1,] 1975 1982
[2,] 1982 1984
[3,] 1984 1991
[4,] 1991 2016
[5,] 2016 2018
[6,] 2018 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1 -31.7150  -31.9320  -31.4970
slope2 387.5900  384.1400  391.0600
slope3 -20.5460  -20.6930  -20.3990
slope4   2.6101    2.5896    2.6306
slope5  54.2150   53.6160   54.8170
slope6 -43.8420  -44.0100  -43.6730
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-4.061761e-02  9.334806e-05 -4.080057e-02 -4.043465e-02
attr(,"wrong.se")
[1] 0.0003590695

=======================
Row: 28
Race: Black
Sex: Female
Site: Arm (shoulder to finger)
Num_Breakpoints: 5
Joinpoint_Years: 1981, 1985, 1996, 2005, 2014
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 1985
[3,] 1985 1996
[4,] 1996 2005
```

```
[5,] 2005 2014
[6,] 2014 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1 -15.8150  -15.9480  -15.6820
slope2  17.0660   16.3920   17.7450
slope3  -8.3661   -8.4210   -8.3111
slope4  10.4840   10.3640   10.6040
slope5   2.3380    2.2831    2.3930
slope6  -6.3846   -6.4356   -6.3336
AAPC:
         Est.        St.Err     CI(95%).l     CI(95%).u
-1.437488e-02  6.303186e-05 -1.449842e-02 -1.425134e-02
attr(,"wrong.se")
[1] 0.000329153

=======================
Row: 29
Race: Black
Sex: Female
Site: Leg (hip to toe)
Num_Breakpoints: 7
Joinpoint_Years: 1978, 1987, 2002, 2006, 2014, 2017, 2020
Segment_Years:
     [,1] [,2]
[1,] 1975 1978
[2,] 1978 1987
[3,] 1987 2002
[4,] 2002 2006
[5,] 2006 2014
[6,] 2014 2017
[7,] 2017 2020
[8,] 2020 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  13.5420   13.2420   13.8430
slope2   1.4084    1.3761    1.4407
slope3  -4.4307   -4.4466   -4.4149
slope4  18.7030   18.4760   18.9300
slope5  -4.1940   -4.2205   -4.1675
slope6 -18.1890  -18.3650  -18.0130
slope7  12.8610   12.4260   13.2970
slope8  36.1230   35.9180   36.3280
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
8.872657e-03 4.382217e-05 8.786767e-03 8.958547e-03
attr(,"wrong.se")
[1] 0.0001765194

=======================
Row: 30
```

```
Race: White
Sex: Male and female
Site: All sites
Num_Breakpoints: 2
Joinpoint_Years: 1981, 2009
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 2009
[3,] 2009 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1 5.61190   5.60770   5.61600
slope2 3.25430   3.25410   3.25460
slope3 0.77341   0.77282   0.77399
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.821649e-02 1.424970e-06 2.821370e-02 2.821928e-02
attr(,"wrong.se")
[1] 2.614306e-06

=======================
Row: 31
Race: White
Sex: Male and female
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1982, 1996, 2002, 2009, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1982
[2,] 1982 1996
[3,] 1996 2002
[4,] 2002 2009
[5,] 2009 2016
[6,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  3.58510   3.57940   3.59070
slope2  3.98480   3.98290   3.98670
slope3  2.88940   2.88240   2.89640
slope4  4.26800   4.26420   4.27180
slope5  1.35670   1.35340   1.35990
slope6 -0.10528  -0.10833  -0.10223
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.841881e-02 2.865514e-06 2.841319e-02 2.842442e-02
attr(,"wrong.se")
[1] 7.758794e-06

=======================
```

```
Row: 32
Race: White
Sex: Male and female
Site: Trunk
Num_Breakpoints: 5
Joinpoint_Years: 1980, 1987, 2007, 2018, 2020
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1987
[3,] 1987 2007
[4,] 2007 2018
[5,] 2018 2020
[6,] 2020 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  7.5944    7.5845    7.6042
slope2  3.8754    3.8716    3.8791
slope3  2.8999    2.8992    2.9006
slope4  1.8727    1.8712    1.8743
slope5 -5.1695   -5.1784   -5.1605
slope6  4.8353    4.8158    4.8547
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.915179e-02 2.966673e-06 2.914597e-02 2.915760e-02
attr(,"wrong.se")
[1] 7.63656e-06

=======================
Row: 33
Race: White
Sex: Male and female
Site: Arm (shoulder to finger)
Num_Breakpoints: 5
Joinpoint_Years: 1980, 1993, 2000, 2007, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1993
[3,] 1993 2000
[4,] 2000 2007
[5,] 2007 2016
[6,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  6.30030   6.29170   6.30880
slope2  3.33170   3.32960   3.33370
slope3  3.68190   3.67670   3.68720
slope4  3.89280   3.88920   3.89640
slope5  2.33200   2.32990   2.33410
slope6 -0.58949  -0.59226  -0.58672
```

```
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.014304e-02 3.001108e-06 3.013716e-02 3.014892e-02
attr(,"wrong.se")
[1] 7.365739e-06

=======================
Row: 34
Race: White
Sex: Male and female
Site: Leg (hip to toe)
Num_Breakpoints: 6
Joinpoint_Years: 1980, 1982, 1993, 1996, 2008, 2019
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1982
[3,] 1982 1993
[4,] 1993 1996
[5,] 1996 2008
[6,] 2008 2019
[7,] 2019 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  2.06330   2.05200   2.07460
slope2 13.91600  13.86500  13.96800
slope3  1.18330   1.18060   1.18600
slope4  5.54850   5.53670   5.56030
slope5  2.58950   2.58740   2.59160
slope6  0.94736   0.94579   0.94892
slope7 -1.81900  -1.83160  -1.80640
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.076170e-02 3.514749e-06 2.075481e-02 2.076858e-02
attr(,"wrong.se")
[1] 1.24792e-05

=======================
Row: 35
Race: White
Sex: Male
Site: All sites
Num_Breakpoints: 6
Joinpoint_Years: 1983, 1996, 2003, 2005, 2016, 2020
Segment_Years:
     [,1] [,2]
[1,] 1975 1983
[2,] 1983 1996
[3,] 1996 2003
[4,] 2003 2005
[5,] 2005 2016
```

```
[6,] 2016 2020
[7,] 2020 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  5.7594    5.7558    5.7630
slope2  4.0193    4.0182    4.0204
slope3  2.5929    2.5904    2.5954
slope4  6.6447    6.6274    6.6620
slope5  1.8939    1.8927    1.8951
slope6 -1.8228   -1.8271   -1.8186
slope7  2.9818    2.9745    2.9890
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.094306e-02 1.916297e-06 3.093930e-02 3.094681e-02
attr(,"wrong.se")
[1] 6.190085e-06

=======================
Row: 36
Race: White
Sex: Male
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1994, 2003, 2005, 2010, 2020
Segment_Years:
     [,1] [,2]
[1,] 1975 1994
[2,] 1994 2003
[3,] 2003 2005
[4,] 2005 2010
[5,] 2010 2020
[6,] 2020 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1 5.11380   5.11230   5.11530
slope2 2.59720   2.59330   2.60110
slope3 8.46160   8.42730   8.49600
slope4 3.53110   3.52450   3.53770
slope5 0.18969   0.18768   0.19171
slope6 1.58510   1.57190   1.59820
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.362661e-02 2.620028e-06 3.362148e-02 3.363175e-02
attr(,"wrong.se")
[1] 1.0028e-05

=======================
Row: 37
Race: White
Sex: Male
Site: Trunk
```

```
Num_Breakpoints: 5
Joinpoint_Years: 1986, 1991, 1996, 2006, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1986
[2,] 1986 1991
[3,] 1991 1996
[4,] 1996 2006
[5,] 2006 2016
[6,] 2016 2022
APC:
          Est. CI(95%).l CI(95%).u
slope1  5.8485    5.8452    5.8517
slope2  1.6010    1.5949    1.6071
slope3  3.9114    3.9006    3.9222
slope4  2.7517    2.7497    2.7536
slope5  1.7156    1.7133    1.7178
slope6 -1.1985   -1.2016   -1.1955
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.677434e-02 2.462738e-06 2.676951e-02 2.677916e-02
attr(,"wrong.se")
[1] 8.21009e-06

=======================
Row: 38
Race: White
Sex: Male
Site: Arm (shoulder to finger)
Num_Breakpoints: 5
Joinpoint_Years: 1980, 1987, 1994, 2006, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1987
[3,] 1987 1994
[4,] 1994 2006
[5,] 2006 2016
[6,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  6.17150   6.15410   6.18880
slope2  5.12450   5.11790   5.13120
slope3  4.40480   4.39820   4.41140
slope4  4.22630   4.22400   4.22860
slope5  1.95480   1.95200   1.95760
slope6 -0.64819  -0.65191  -0.64447
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.405187e-02 4.720060e-06 3.404262e-02 3.406112e-02
attr(,"wrong.se")
```

```
[1] 1.217627e-05

=======================
Row: 39
Race: White
Sex: Male
Site: Leg (hip to toe)
Num_Breakpoints: 6
Joinpoint_Years: 1989, 2000, 2002, 2008, 2011, 2015
Segment_Years:
     [,1] [,2]
[1,] 1975 1989
[2,] 1989 2000
[3,] 2000 2002
[4,] 2002 2008
[5,] 2008 2011
[6,] 2011 2015
[7,] 2015 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  3.03220   3.02780   3.03660
slope2  4.18090   4.17610   4.18580
slope3 -3.35980  -3.41610  -3.30350
slope4  5.43700   5.42920   5.44480
slope5 -2.59090  -2.61570  -2.56600
slope6  5.31500   5.28950   5.34050
slope7 -0.83013  -0.83512  -0.82514
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.507230e-02 4.444148e-06 2.506359e-02 2.508101e-02
attr(,"wrong.se")
[1] 2.120256e-05

=======================
Row: 40
Race: White
Sex: Female
Site: All sites
Num_Breakpoints: 5
Joinpoint_Years: 1982, 1993, 1999, 2005, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1982
[2,] 1982 1993
[3,] 1993 1999
[4,] 1999 2005
[5,] 2005 2016
[6,] 2016 2022
APC:
            Est. CI(95%).l CI(95%).u
slope1 4.9671000 4.9626000   4.97160
```

```
slope2 1.9550000 1.9534000   1.95660
slope3 3.5009000 3.4956000   3.50620
slope4 3.2728000 3.2699000   3.27560
slope5 1.8812000 1.8800000   1.88250
slope6 0.0051692 0.0023684   0.00797
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.462345e-02 2.017954e-06 2.461949e-02 2.462740e-02
attr(,"wrong.se")
[1] 5.739541e-06

=======================
Row: 41
Race: White
Sex: Female
Site: Head and Neck
Num_Breakpoints: 5
Joinpoint_Years: 1980, 1987, 1995, 1998, 2009
Segment_Years:
     [,1] [,2]
[1,] 1975 1980
[2,] 1980 1987
[3,] 1987 1995
[4,] 1995 1998
[5,] 1998 2009
[6,] 2009 2022
APC:
         Est. CI(95%).l CI(95%).u
slope1 3.4722    3.4528    3.4917
slope2 1.3616    1.3533    1.3699
slope3 1.9428    1.9335    1.9521
slope4 5.5567    5.5240    5.5894
slope5 2.2429    2.2396    2.2461
slope6 1.0366    1.0341    1.0390
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.062830e-02 5.470946e-06 2.061758e-02 2.063903e-02
attr(,"wrong.se")
[1] 1.876408e-05

=======================
Row: 42
Race: White
Sex: Female
Site: Trunk
Num_Breakpoints: 5
Joinpoint_Years: 1978, 1989, 1992, 2005, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1978
[2,] 1978 1989
```

```
[3,] 1989 1992
[4,] 1992 2005
[5,] 2005 2016
[6,] 2016 2022
APC:
            Est. CI(95%).l CI(95%).u
slope1 11.423000 11.397000 11.448000
slope2  4.011300  4.006600  4.015900
slope3  0.074709  0.049715  0.099709
slope4  3.591300  3.589300  3.593400
slope5  2.306600  2.303900  2.309400
slope6  0.163060  0.158840  0.167270
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
3.129782e-02 5.190834e-06 3.128765e-02 3.130800e-02
attr(,"wrong.se")
[1] 1.32766e-05

=======================
Row: 43
Race: White
Sex: Female
Site: Arm (shoulder to finger)
Num_Breakpoints: 5
Joinpoint_Years: 1981, 1993, 2001, 2010, 2016
Segment_Years:
     [,1] [,2]
[1,] 1975 1981
[2,] 1981 1993
[3,] 1993 2001
[4,] 2001 2010
[5,] 2010 2016
[6,] 2016 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  6.17280   6.16140   6.18420
slope2  1.82230   1.81920   1.82540
slope3  3.56680   3.56270   3.57090
slope4  3.14490   3.14150   3.14830
slope5  2.86230   2.85490   2.86960
slope6 -0.55572  -0.55977  -0.55168
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
2.716808e-02 4.067018e-06 2.716011e-02 2.717605e-02
attr(,"wrong.se")
[1] 1.076226e-05

=======================
Row: 44
Race: White
Sex: Female
```

```
Site: Leg (hip to toe)
Num_Breakpoints: 6
Joinpoint_Years: 1977, 1984, 1994, 1997, 2006, 2019
Segment_Years:
     [,1] [,2]
[1,] 1975 1977
[2,] 1977 1984
[3,] 1984 1994
[4,] 1994 1997
[5,] 1997 2006
[6,] 2006 2019
[7,] 2019 2022
APC:
           Est. CI(95%).l CI(95%).u
slope1  1.93720   1.90730   1.96700
slope2  5.13790   5.12880   5.14710
slope3  0.82134   0.81781   0.82488
slope4  6.19260   6.17090   6.21440
slope5  2.82750   2.82410   2.83090
slope6  0.89265   0.89115   0.89414
slope7 -2.18020  -2.19530  -2.16510
AAPC:
        Est.       St.Err    CI(95%).l    CI(95%).u
1.970128e-02 4.674599e-06 1.969212e-02 1.971045e-02
attr(,"wrong.se")
[1] 1.349182e-05
```